

FH Salzburg
Information Technologies
and Digitalisation

Software and Process Notations

Laboratory – Introduction and Course Organization

competent
relevant
sustainable

Teaching Team

Theory

- FH-Prof. Dr. Christian Neureiter

Laboratory

- FH-Prof. Dr. Christoph Binder
- DI Katharina Polanec
- DI Markus Michael Peter
- DI Jounes-Alexander Gross

Tutorial

- Simon Eschlberger



Goals of this course



Theory

- Understand the complexity of software-intense systems
- Basic knowledge on language theory, notations, and models
- Basic knowledge on Enterprise Architecture
  - Frameworks and Notations
- Basics of Software Language Engineering
  - Domain Specific Languages (DSLs)
  - Language design on basis of Metamodels
  - Language implementation with UML profiles

Application of Theory

- Systems engineering and modeling
- Utilization of CASE tools
  - Sparx Systems „Enterprise Architect“
- Development of Enterprise Architectures
  - Organizational and Process description
- Domain Specific Languages (DSLs)
  - Specification
  - Implementation as UML-Profile

Grading

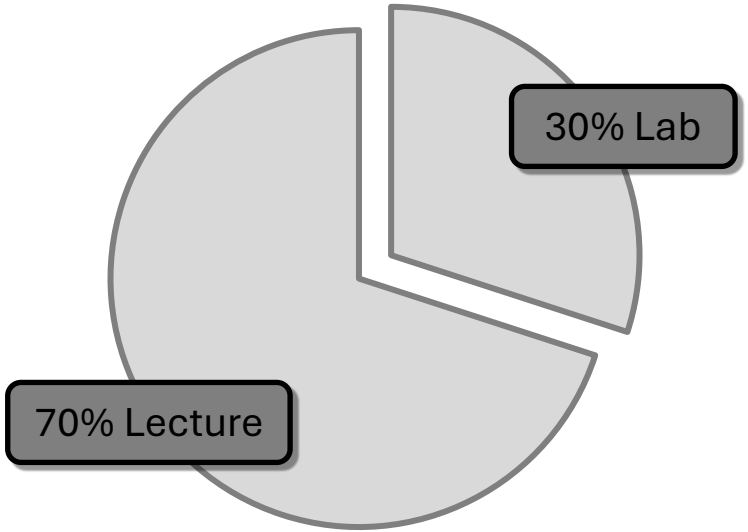


Your grade consists of two parts:

- **Lecture**
  - *Written multiple-choice exam: 70%*
- **Lab**
  - *Submitted exercises: 30%*

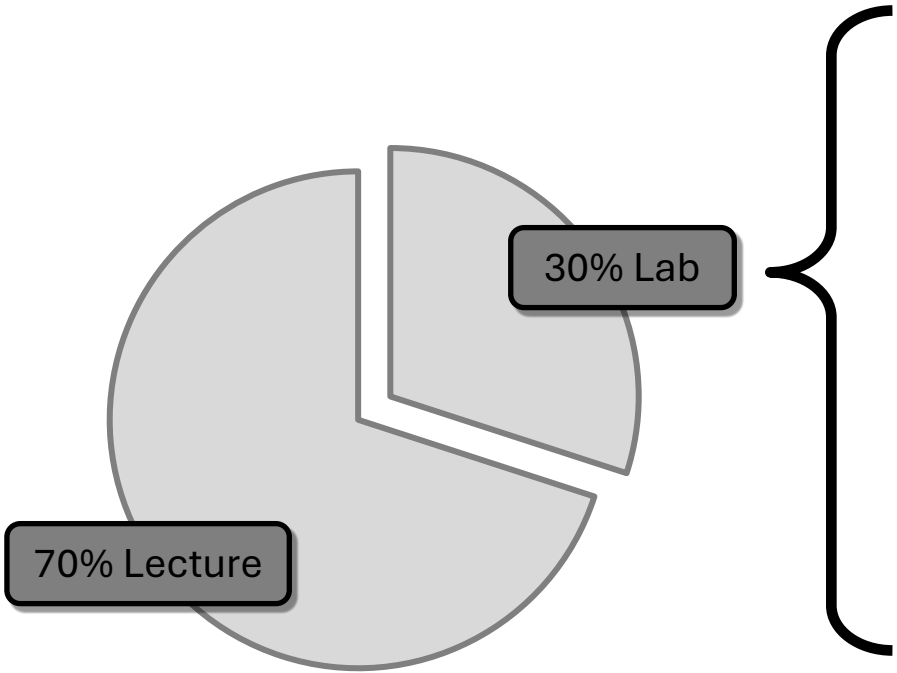
Grades and points

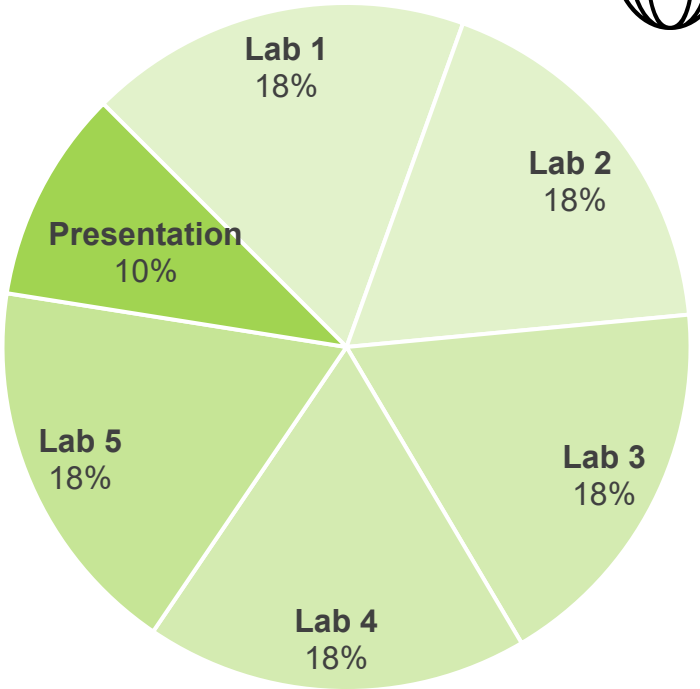
- **1** – *Sehr gut*: **[87,5 - 100]**
- **2** – *Gut*: **[75 - 87,5)**
- **3** – *Befriedigend*: **[62,5 - 75)**
- **4** – *Genügend*: **[50 - 62,5)**
- **5** – *Nicht genügend*: **[0 - 50)**



***Both parts** need to be positive to complete the course!*

Grading


30% Lab
70% Lecture


Lab 1
18%
Lab 2
18%
Presentation
10%
Lab 5
18%
Lab 3
18%
Lab 4
18%

Additional Information



Grading of the laboratory

- Submitted examples:
  - Level of detail
  - Appearance of model
  - Plausibility
  - Effort
- Every laboratory is equally weighted

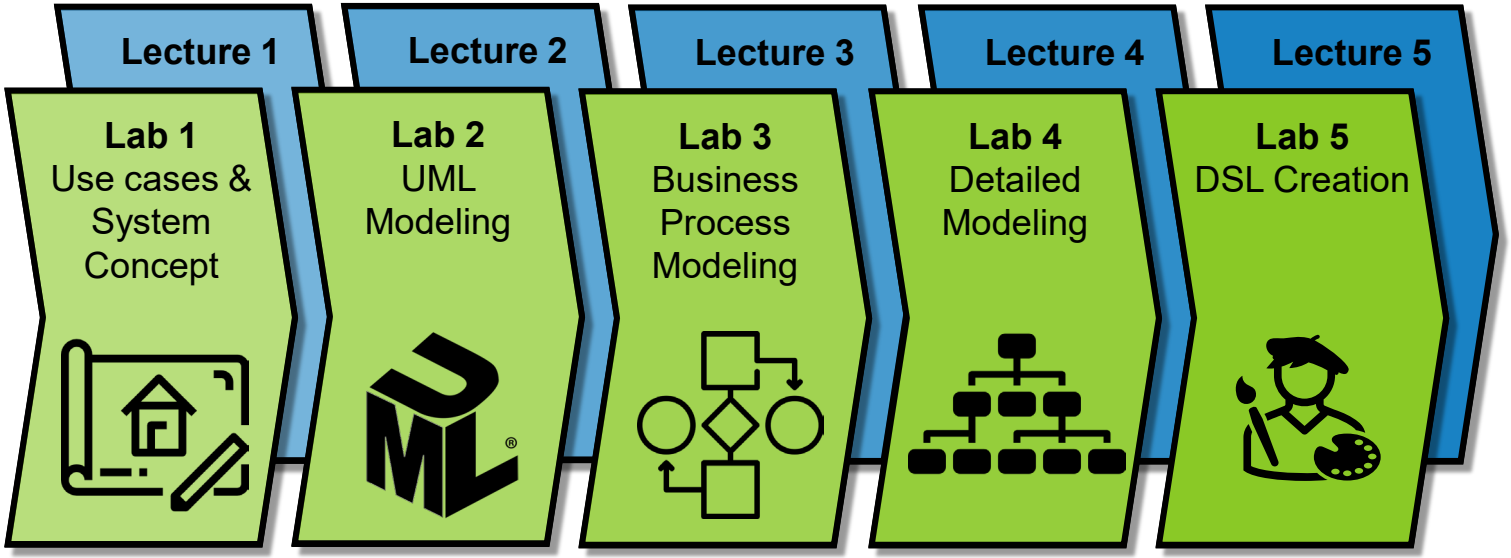
Presence

- At least 75% attendance
- Ability of independent work is required
- Working in teams is encouraged

In case of questions

- Moodle Forum
- Your questions will be answered as soon as possible by:
  - the lecturer team
  - your colleagues from other groups/courses (Hint: we do appreciate horizontal knowledge-sharing and learning ☺)
- Questions regarding the content will not be answered via e-mail

Course Organization and Context





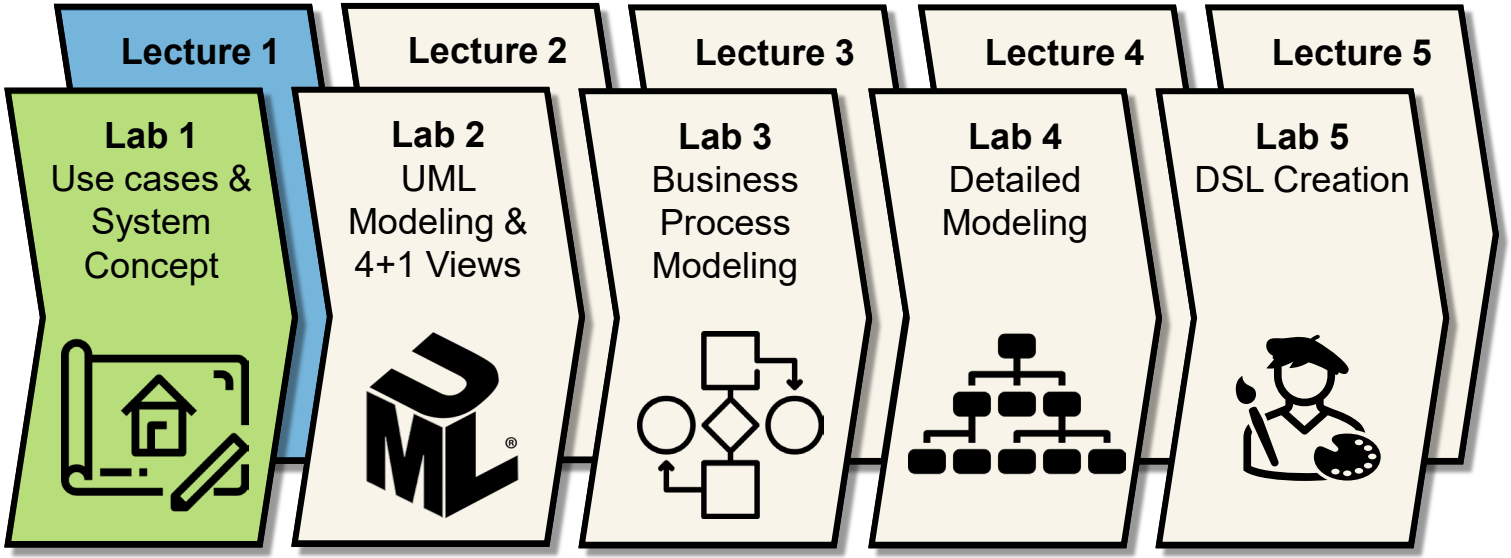
FH Salzburg
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Software and Process Notations

Laboratory 1 – Use Cases & System Concept

competent
relevant
sustainable

Where are we?


Lecture 1
Lecture 2
Lecture 3
Lecture 4
Lecture 5
Lab 1
Use cases & System Concept
Lab 2
UML Modeling & 4+1 Views
Lab 3
Business Process Modeling
Lab 4
Detailed Modeling
Lab 5
DSL Creation



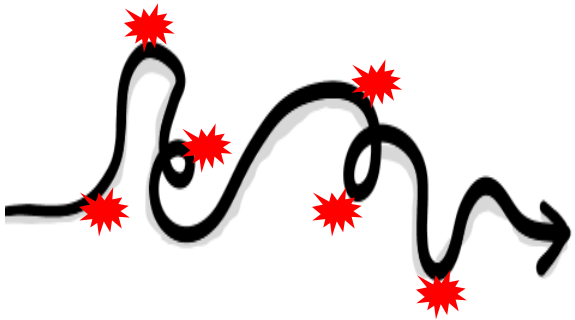
BACKGROUND INFORMATION

Software and Process Notations – LAB 1

Background Story

Plan



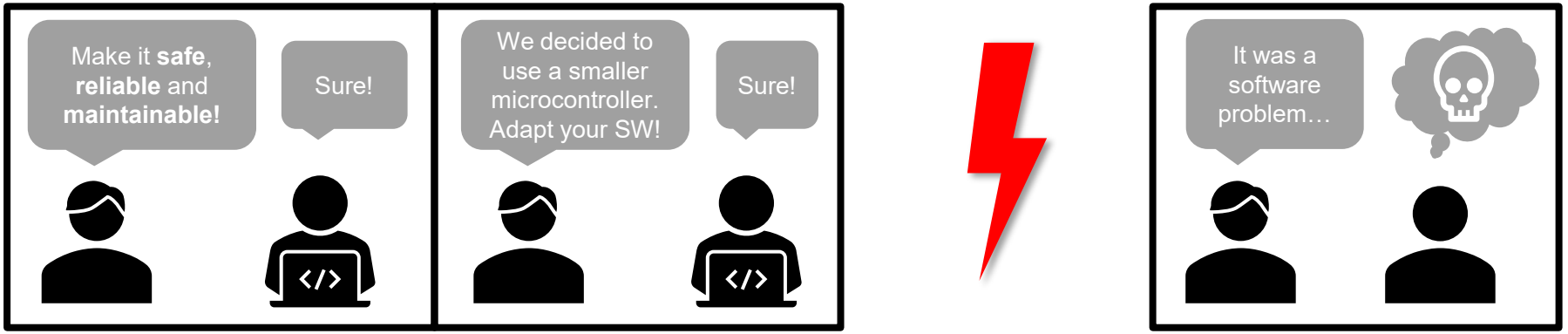


Realization



What is happening here?

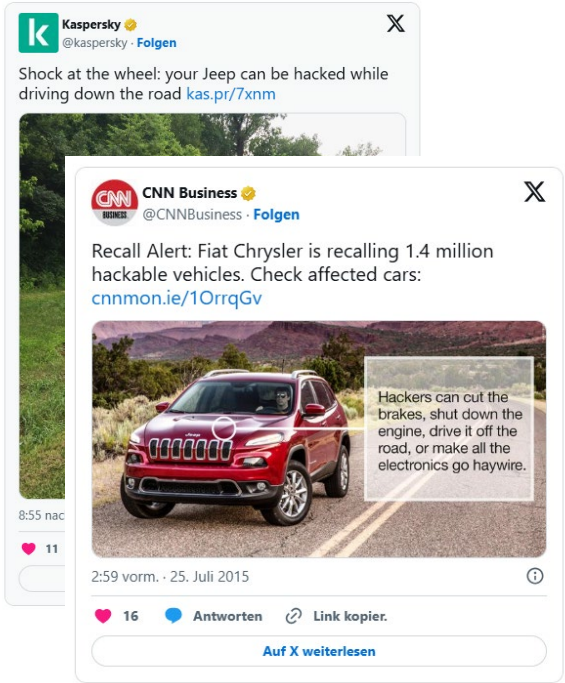
Background Story



When things go wrong: Jeep Cherokee Hack

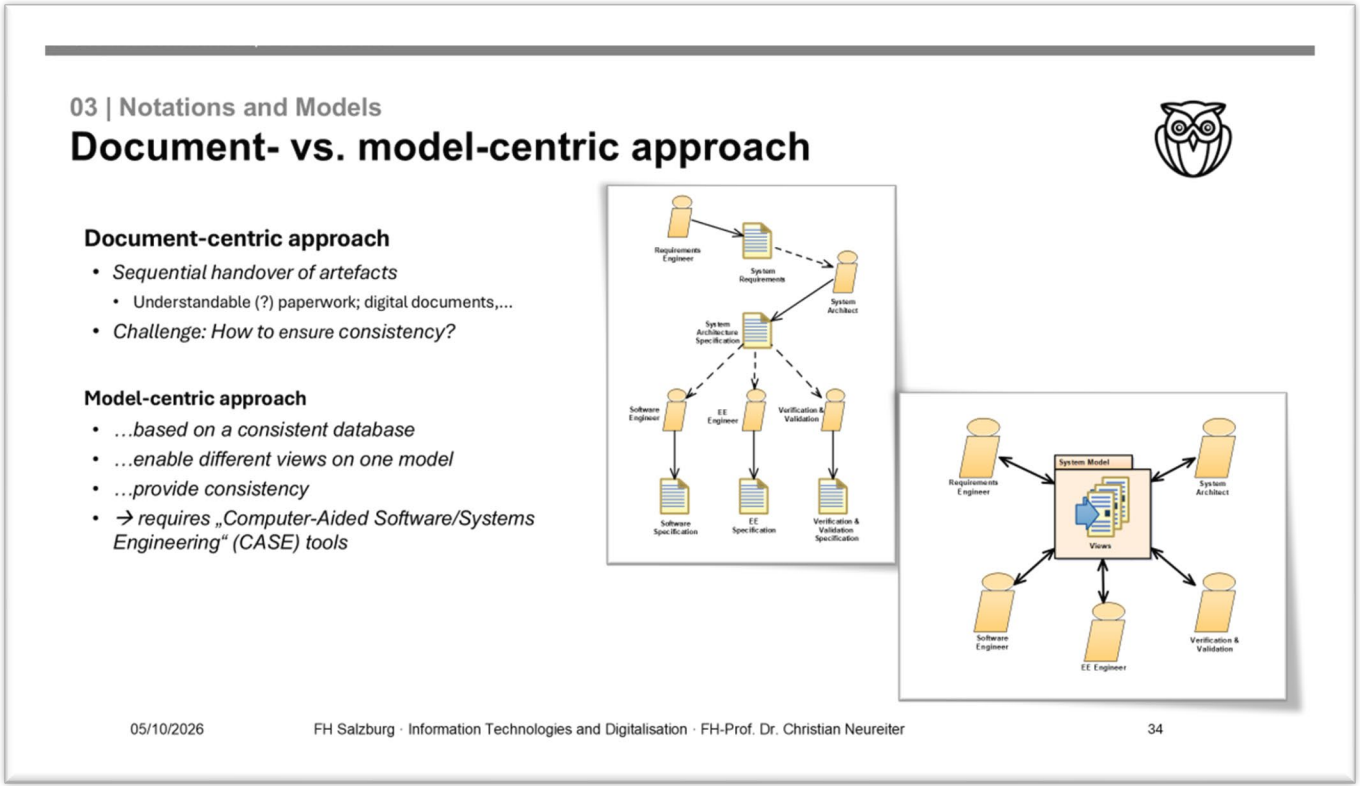
System context: Connected vehicle with wireless infotainment system (Uconnect).
What happened: Security researchers remotely took control of a Jeep over the internet — steering, braking, acceleration.

Poor interface isolation & network architecture: The infotainment system had insufficient isolation from safety-critical ECUs; network segmentation and gateway policies were deficient.
Requirements omission for security: Cybersecurity requirements (threat models, secure update mechanisms, remote access controls) were missing or not given equal priority to functional requirements during system design.

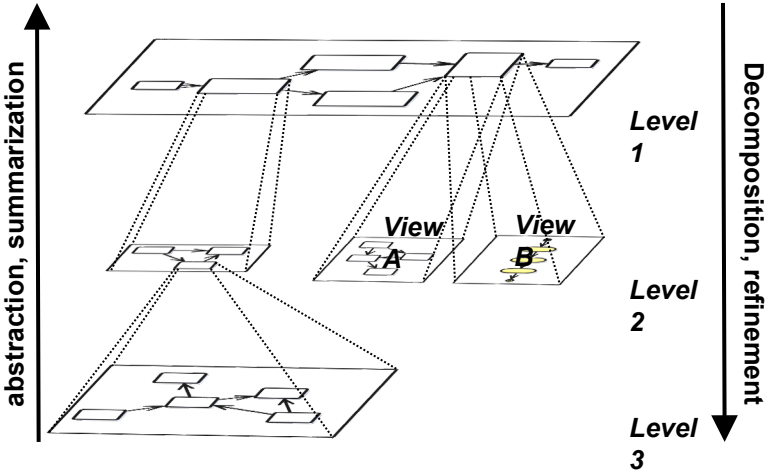


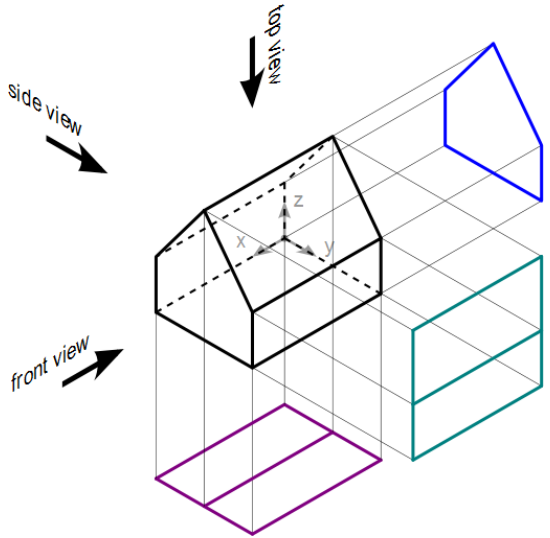
https://www.kaspersky.de/blog/blackhat-jeep-cherokee-hack-explained/5940/

Recap


03 | Notations and Models
Document- vs. model-centric approach
Document-centric approach
Sequential handover of artefacts
Understandable (?) paperwork; digital documents,...
Challenge: How to ensure consistency?
Model-centric approach
…based on a consistent database
…enable different views on one model
…provide consistency
→ requires „Computer-Aided Software/Systems Engineering“ (CASE) tools
Requirements Engineer
System Requirements
System Architect
System Architecture Specification
Software Engineer
EE Engineer
Verification & Validation
Software Specification
EE Specification
Verification & Validation Specification
System Model
Views
05/10/2026
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Model-Based Software Engineering







EXERCISE INFORMATION

Software and Process Notations – LAB 1

Real-World Scenario: Automatic Valet Parking / Automatic Valet Charging

- *Level 4 autonomy*
- *Drop-off zone*
- *Parking garage*
- *Cameras*
- *On-board sensors of the vehicle*



Automated Valet Parking & Charging



Integration with Payment Systems

- There is no link between the charging service and a payment system, making it cumbersome for users to pay for charging
- **How can payment for charging be made safe, seamless and automatic?**

Compliance and Data Privacy Management

- Cameras localize vehicles, but they also record people and licence plates.
- **Where is personal data anonymized, and how is access to raw footage controlled and logged?**

Incident Monitoring and Anomaly Detection

- The current system lacks a dedicated mechanism for monitoring incidents or detecting anomalies, which can lead to unresolved issues that disrupt the charging service.
- **How can the system be improved to monitor charging incidents and detect anomalies in real time to ensure continuous and safe operation?**

User Identity and access management

- There is no robust system in place to manage user identities and access rights, leading to potential security vulnerabilities and unauthorized access to sensitive data.
- **How can a secure identity and access management system be integrated to protect user information and ensure that only authorized users can access the charging service?**



Use Case Description - Template



A use case is a specification of behavior.

Use Case Name:	
Description:	
Preconditions:	
Postconditions:	
Frequency of Use:	
Normal Course of Events:	
Alternative Course:	
Exceptions:	
Includes:	
Special Requirement:	
Assumptions:	

Template based on IEC 62559

Use Case: Pizza



Actor:	PizzaBot3000
Description:	Users can browse the available pizzas, place an order, and pay directly using their phone. Once an order is confirmed, PizzaBot3000 prepares and bakes the pizza on the automated pizza-baking street. After the pizza is freshly baked, a PizzaBot robot transports the order directly to the customer's designated pickup or delivery location.
Preconditions:	User is logged in, craving pizza, and has a valid delivery address; PizzaBot3000 is operational.
Postconditions:	Pizza delivered hot and fresh; user satisfied; PizzaBot3000 preps for the next order.
Frequency of Use:	As often as hunger strikes—typically around meal times, midnight cravings, and during Netflix binges.
Normal Course of Events:	User orders, PizzaBot3000 bakes, DeliveryDrone420 flies, pizza arrives, and user devours it with joy.
Alternative Course:	User cancels the order mid-bake, PizzaBot3000 gracefully redirects the pizza to the nearest hungry neighbor. Or the void…
Exceptions:	Oven meltdown or rogue AI; PizzaBot3000 declares independence, starts a food truck, and offers the user a discount code.
Includes:	Order processing, pizza baking, real-time tracking, and cancellation options with cheesy bot responses.
Special Requirement:	System must comply with dietary restrictions and ensure pineapple on pizza is an opt-in feature only.
Assumptions:	User has a stable internet connection, a functional delivery address, and isn't allergic to gluten—or robots.

Excercise: System Concept

1) **Get together in groups of 3** *(You will work in these groups for the rest of the lecture)*
2) **Based on the scenario develop a use case for your group and fill out the template.**
3) **Sketch your system for an investor in the following perspectives:**



Objective/Physical-Based (Objective)
"What physical components or artifacts make up the system, and how do they interact?"



Process-Based (Functional)
"What sequence of actions or steps does the system follow to achieve its goals?"



Time-Based (Temporal)
"How does the System behave over time?"

Assignment



Deliverables

1) Use Case template [PDF]
2) Three sketches of your system [PDF]

Deadline

1) According to Moodle

Preparation for next Lab

1) Read & Understand the following article:
   https://medium.com/@rushikeshdhurat/model-driven-engineering-ff6e85f76e19

➔ **There will be a quick multiple choice quiz next Lab**

